

How available information shapes denoising and computation

I: participants

woman, man, blue cup

R: interaction roles

giver, receiver, object

V: visual realization

hand and cup positions

A. What does clearer role information contribute to visual prediction?

Hold participants, visual-target noise, and transformer weights fixed.

Noisier roles R

woman ← who gives? → man

Denoise the same hand/cup states V

Clearer roles R

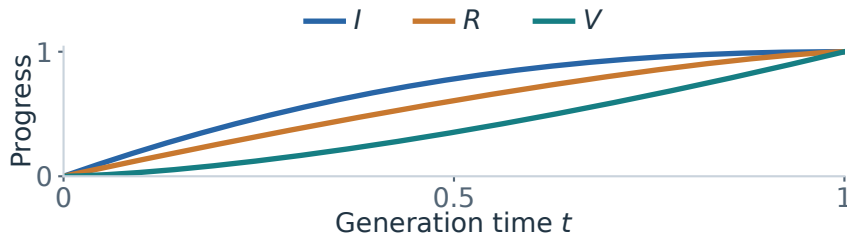
woman — gives cup —→ man

Denoise the same hand/cup states V

Compare visual denoising losses to measure the contribution of R.

Role labels are withheld from the prompt in this training comparison.

B. Learn component schedules with the transformer



Earlier R can help V, but R must be predicted with less visual information.

All components continue developing.

C. Identify the information paths and computations that can be reused

I: participants

R: noisy roles

V: noisy visual

Attention head

Shared MLP

Map to V

$g_m(\mathbf{u})$

Predict V

Map to R

Predict R

Gate $g_m(\mathbf{u})$ changes this contribution with component noise levels.

Dashed: reuse the predictor for roles, with task-specific input/output maps.

Illustrative components, schedules, and module roles; these are learned in the project.